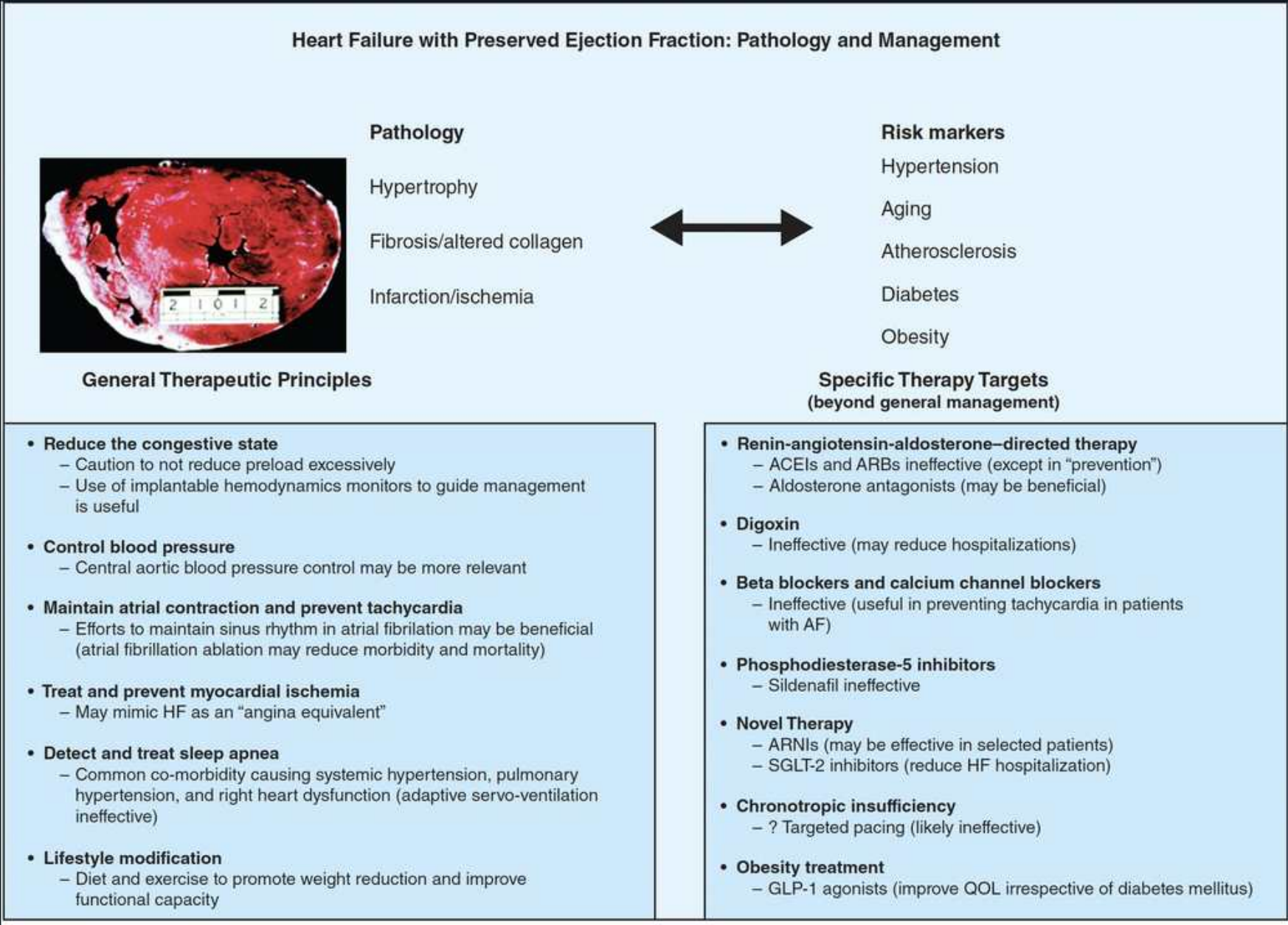


Heart Failure — Pathophysiology correlation of general therapeutic HF



1. HFpEF pathology: hypertrophy

WHAT YOU SEE

Pathology list 'Hypertrophy' next to heart specimen

WHAT IT ENCODES

HFpEF substrate is concentric LV hypertrophy from chronic pressure overload (HTN, AS): wall thickening with a small, stiff, non-compliant cavity. Stiffness shifts the diastolic pressure-volume curve up and left, so filling pressures rise steeply for small volume increases (impaired relaxation + ↑passive stiffness). EF is preserved (≥50%) but stroke volume is filling-limited. On the Frank-Starling frame, HFpEF is a problem of getting blood IN (diastole), whereas HFrEF is a problem of pumping blood OUT (systole).

BOARD TRAP

WRONG: 'EF is preserved, so cardiac output and prognosis are normal.' HFpEF carries mortality comparable to HFrEF; the limitation is diastolic filling and reduced stroke volume reserve, not contractility.

2. Fibrosis / altered collagen

WHAT YOU SEE

'Fibrosis / altered collagen'

WHAT IT ENCODES

Interstitial and perivascular fibrosis with a shift toward stiffer type I collagen increases passive myocardial stiffness, the dominant driver of diastolic dysfunction in HFpEF. Titin hypophosphorylation and microvascular inflammation (driven by HTN, DM, obesity) add to the stiffness. This is why anti-fibrotic/RAAS strategies were trialed; MRAs (spironolactone) blunt fibrosis and may help selected HFpEF.

BOARD TRAP

WRONG: 'Diastolic dysfunction in HFpEF is purely an active relaxation (lusitropy) defect.' Passive stiffness from fibrosis/collagen is the major contributor; that is why simply slowing the heart is not curative.

3. Infarction / ischemia substrate

WHAT YOU SEE

'Infarction / ischemia'

WHAT IT ENCODES

Prior MI and ongoing ischemia stiffen the ventricle (scar) and acutely impair ATP-dependent relaxation, precipitating flash pulmonary edema even with preserved EF. Demand ischemia raises LV diastolic pressure abruptly. This is why HF can behave as an 'anginal equivalent' and why revascularization/anti-ischemic therapy is part of HFpEF care.

BOARD TRAP

WRONG: 'Ischemia only matters in HFrEF.' Transient ischemia is a classic trigger of acute HFpEF (flash pulmonary edema with normal EF on later echo).

4. Risk markers: HTN, aging

WHAT YOU SEE

Right column 'Hypertension', 'Aging'

WHAT IT ENCODES

Hypertension and advancing age are the two strongest HFpEF risk markers: chronic afterload drives hypertrophy/fibrosis, and aging adds intrinsic myocardial and arterial stiffening (↑central aortic/pulse-wave pressure). The typical board patient is an older woman with HTN, dyspnea on exertion, preserved EF, and elevated NT-proBNP. Aggressive BP control is the single highest-yield preventive intervention.

BOARD TRAP

WRONG: 'HFpEF mainly affects young men with dilated hearts.' The prototype is an elderly hypertensive woman, often obese/diabetic, with a non-dilated stiff ventricle.

5. Risk markers: atherosclerosis, DM, obesity

WHAT YOU SEE

'Atherosclerosis', 'Diabetes', 'Obesity'

WHAT IT ENCODES

Atherosclerosis, diabetes, and obesity create the systemic inflammatory/metabolic milieu of HFpEF (coronary microvascular dysfunction, advanced glycation crosslinks, plasma volume expansion). Obesity-phenotype HFpEF has high filling pressures from volume overload and pericardial restraint. These comorbidities, not contractile failure, define the syndrome — and they explain why SGLT2i and GLP-1 agonists (metabolic drugs) became HFpEF therapies.

BOARD TRAP

WRONG: 'Treating HFpEF means optimizing contractility.' There is no inotrope benefit; management targets the comorbidities (BP, weight, glucose, ischemia, AF) and congestion.

6. Reduce congestion (careful preload)

WHAT YOU SEE

'Reduce the congestive state — caution not to reduce preload excessively'

WHAT IT ENCODES

The stiff HFpEF ventricle sits on the steep part of the Frank-Starling/pressure-volume curve, so it is exquisitely preload-dependent: loop diuretics relieve congestion, but over-diuresis drops filling and cardiac output, causing hypotension and pre-renal AKI. Diurese to euvolemia and reassess; invasive/implantable hemodynamic monitoring (e.g., CardioMEMS) can guide titration.

BOARD TRAP

WRONG: 'Aggressive, rapid diuresis is always safe in HFpEF.' Excessive preload reduction in a stiff ventricle precipitates hypotension and renal injury — titrate gently to euvolemia.

7. Control blood pressure

WHAT YOU SEE

'Control blood pressure — central aortic pressure more relevant'

WHAT IT ENCODES

BP control is the cornerstone of HFpEF: lowering afterload reduces the hypertrophy/fibrosis stimulus and acutely lowers filling pressures. Central aortic pressure (pulse-wave reflection) correlates with LV load better than brachial cuff in the stiff-vessel elderly. Target guideline BP (generally <130/80); any first-line agent that also treats a comorbidity (ACEi/ARB for DM, MRA for fibrosis) is reasonable.

BOARD TRAP

WRONG: 'Brachial cuff pressure fully captures the ventricular load in HFpEF.' In stiff arteries the central aortic systolic pressure is the more relevant load on the LV.

8. Maintain sinus / treat AF

WHAT YOU SEE

'Maintain atrial contraction and prevent tachycardia'

WHAT IT ENCODES

A stiff ventricle relies on the atrial kick (up to ~25–30% of filling) and on a long diastole, so loss of sinus rhythm or tachycardia abruptly worsens HFpEF — new AF with RVR is a classic decompensation trigger. Maintain sinus rhythm where feasible and rate-control AF; slowing rate lengthens diastolic filling time. AF ablation may reduce morbidity in selected patients.

BOARD TRAP

WRONG: 'Rate vs rhythm doesn't matter much in HFpEF because EF is normal.' Loss of atrial kick and tachycardia critically impair filling of a non-compliant ventricle and commonly precipitate decompensation.

9. Treat ischemia / sleep apnea

WHAT YOU SEE

'Treat myocardial ischemia', 'Detect and treat sleep apnea'

WHAT IT ENCODES

Treat reversible drivers: myocardial ischemia (HFpEF as an anginal equivalent, flash edema) and obstructive sleep apnea, a common comorbidity causing nocturnal sympathetic surges, systemic + pulmonary hypertension, and right-heart strain. Screen and treat OSA (CPAP). Note: adaptive servo-ventilation is HARMFUL in HFrEF with central apnea (SERVE-HF ↑mortality) — do not confuse the two.

BOARD TRAP

WRONG: 'Adaptive servo-ventilation should be used for sleep-disordered breathing in heart failure.' ASV increased mortality in HFrEF with predominant central sleep apnea (SERVE-HF) — it is contraindicated there; OSA gets CPAP.

10. RAAS therapy ineffective in HFpEF

WHAT YOU SEE

'ACEi/ARB ineffective (except prevention); aldosterone antagonists may help'

WHAT IT ENCODES

Unlike HFrEF — where neurohormonal RAAS blockade (ACEi/ARB/ARNI) is mortality-reducing standard of care — ACEi/ARB do NOT reduce mortality in established HFpEF (CHARM-Preserved, I-PRESERVE neutral); they remain useful for prevention by treating HTN. MRAs (spironolactone, TOPCAT) may reduce HF hospitalizations in selected HFpEF, plausibly via anti-fibrotic action. Watch hyperkalemia/renal function on MRA + ACEi.

BOARD TRAP

WRONG: 'ACEi/ARB reduce mortality in HFpEF just as in HFrEF.' They are effective in HFrEF but ineffective for outcomes in established HFpEF (only useful for BP/prevention); the partial winner in HFpEF is the MRA.

11. Digoxin / beta-blocker ineffective

WHAT YOU SEE

'Digoxin ineffective', 'Beta blockers & CCB ineffective (useful for AF rate)'

WHAT IT ENCODES

Digoxin is an inotrope/AV-nodal agent and offers no outcome benefit in HFpEF (DIG-PEF: ↔ mortality, no symptom gain). Beta-blockers and non-dihydropyridine CCBs (diltiazem/verapamil) don't improve HFpEF outcomes either, but are useful for rate control of AF (lengthening diastolic filling). Avoid excessive bradycardia, which can lower output in a preload-dependent ventricle.

BOARD TRAP

WRONG: 'Beta-blockers improve survival in HFpEF the way they do in HFrEF.' Their HFpEF role is limited to AF rate control; the mortality benefit of beta-blockers is an HFrEF phenomenon.

12. PDE-5 inhibitor (sildenafil) ineffective

WHAT YOU SEE

'Phosphodiesterase-5 inhibitors — sildenafil ineffective'

WHAT IT ENCODES

PDE-5 inhibitors (sildenafil) were hypothesized to help via NO–cGMP enhancement and pulmonary vasodilation, but RELAX showed NO improvement in exercise capacity or clinical status in HFpEF. They remain a treatment for pulmonary ARTERIAL hypertension (Group 1), not for the post-capillary PH of HFpEF (Group 2).

BOARD TRAP

WRONG: 'Sildenafil treats the pulmonary hypertension of HFpEF.' It failed in HFpEF (RELAX) and is not used for Group 2 (post-capillary) PH; reserve PDE-5i for Group 1 PAH.

13. SGLT2i reduce HF hospitalization

WHAT YOU SEE

'Novel therapy: SGLT2 inhibitors reduce HF hospitalization'

WHAT IT ENCODES

SGLT2 inhibitors (dapagliflozin DELIVER, empagliflozin EMPEROR-Preserved) are the breakthrough HFpEF class: they reduce HF hospitalization and CV death across the full EF spectrum, regardless of diabetes — mechanisms include natriuresis/decongestion, improved myocardial energetics, and reduced inflammation. They are now a guideline Class I/IIa pillar for HFpEF. ARNI (sacubitril-valsartan, PARAGON-HF) narrowly missed significance but may help lower-EF-range HFpEF.

BOARD TRAP

WRONG: 'No drug improves outcomes in HFpEF.' SGLT2 inhibitors reduce HF hospitalizations/CV death in HFpEF regardless of diabetes — they are the class to know.

14. Obesity: GLP-1 agonists

WHAT YOU SEE

'Obesity treatment: GLP-1 agonists improve QOL irrespective of diabetes'

WHAT IT ENCODES

In the obesity phenotype of HFpEF, GLP-1 receptor agonists (semaglutide, STEP-HFpEF) improve symptoms, exercise capacity (6-minute walk), and quality of life with weight loss — benefit seen irrespective of diabetes status. This reframes obesity as a treatable driver, not just a comorbidity. Combine with diet/exercise; weight reduction lowers plasma volume, pericardial restraint, and inflammation.

BOARD TRAP

WRONG: 'GLP-1 agonists only matter for glycemic control and have no role in non-diabetic HFpEF.' In obesity-phenotype HFpEF they improve symptoms and QOL independent of diabetes (STEP-HFpEF).